

1.2: Domain and Range

Definition. (Interval notation)

We can specify the domain and/or range of a function using interval notation:

- Strict inequalities (exclusive) are represented using parenthesis:

$$x > 3 \iff (3, \infty) \qquad x < 9 \iff (-\infty, 9)$$

Note: Parenthesis are **always** used when an interval's end points are infinite

- Nonstrict inequalities (inclusive) are represented using square brackets:

$$7 \leq x \leq 11 \iff [7, 11]$$

- Disjoint intervals can be unioned together:

$$x \neq -2 \iff (-\infty, -2) \cup (-2, \infty)$$

- Any combination of the above can be used:

$$[-1, 1] \cup (2, 3] \cup (4, 5) \cup [6, \infty)$$



Recall that the domain is the set of all *possible* input values.

Example. Find the domain of the following functions (use interval notation): [Graphs](#)

$$f(x) = x - 3$$

$$\text{Domain : } (-\infty, \infty)$$

$$\text{Range : } (-\infty, \infty)$$

$$A = \pi r^2$$

$$\text{Domain : } (-\infty, \infty)$$

$$\text{Range : } [0, \infty)$$

$$y = \sqrt{x-1}$$

$$\text{Domain : } [1, \infty)$$

$$\text{Range : } [0, \infty)$$

$$y = \frac{x+2}{x^2-4}$$

$$\text{Domain : } (-\infty, -2) \cup (-2, 2) \cup (2, \infty)$$

$$\text{Range : } (-\infty, -\frac{1}{4}) \cup (-\frac{1}{4}, 0) \cup (0, \infty)$$

$$y = \sqrt[3]{x-4}$$

$$\text{Domain : } (-\infty, \infty)$$

$$\text{Range : } (-\infty, \infty)$$

$$f(x) = \frac{\sqrt[4]{x^2+4}}{(x+1)\sqrt{x-1}}$$

$$\text{Domain : } (1, \infty)$$

$$\text{Range : } (0, \infty)$$

Example. Identify the domain and range of the following:

$$x(x+1)(x+2)^2, \quad x \geq -2$$

$$\sin(x)$$

$$\text{Domain : } (-\infty, \infty)$$

$$\text{Domain : } (-\infty, \infty)$$

$$\text{Range : } (-0.61968, \infty)$$

$$\text{Range : } [-1, 1]$$

$$\log(x)$$

$$2^x, \quad x \geq -1$$

$$\text{Domain : } (0, \infty)$$

$$\text{Domain : } [-1, \infty)$$

$$\text{Range : } (-\infty, \infty)$$

$$\text{Range : } [\frac{1}{2}, \infty)$$

Definition.

A **piecewise** function is a function with different definitions for different portions of the domain.

Example. Rewrite the following as piecewise functions:

$$|x| = \begin{cases} x, & \text{if } x \geq 0 \\ -x, & \text{if } x < 0 \end{cases}$$

$$\frac{x}{|x|} = \begin{cases} \frac{x}{x}, & \text{if } x > 0 \\ -\frac{x}{x}, & \text{if } x < 0 \end{cases}$$

$$= \begin{cases} 1, & \text{if } x > 0 \\ -1, & \text{if } x < 0 \end{cases}$$

$$|x-1| + |4-x| = \begin{cases} -2x+5, & \text{if } x \leq 1 \\ 3, & \text{if } 1 < x \leq 4 \\ 2x-5, & \text{if } x > 4 \end{cases}$$

Example. Graph the following piecewise function:

[Graphs](#)

$$f(x) = \begin{cases} x^2, & \text{if } x \leq 1 \\ 3, & \text{if } 1 < x \leq 2 \\ x, & \text{if } x > 2 \end{cases}$$

